```python
import os
import shutil
from PIL import Image, ImageFilter
from pathlib import Path
import matplotlib.pyplot as plt
import numpy as np
from skimage import data

output_dir = "./style_transfer_dataset"
img_size = (256, 256)
train_split = 0.9

sample_images = {
    "camera": data.camera(),
    "chelsea": data.chelsea(),
    "astronaut": data.astronaut(),
    "coffee": data.coffee(),
    "rocket": data.rocket()
}

def create_dir_structure():
    for split in ["trainA", "trainB", "testA", "testB"]:
        os.makedirs(os.path.join(output_dir, split), exist_ok=True)

def save_sample_images():
    print("Saving sample images...")
    os.makedirs("./raw_dataset/photos", exist_ok=True)
    os.makedirs("./raw_dataset/paintings", exist_ok=True)
    for i, (name, img) in enumerate(sample_images.items()):
        photo = Image.fromarray(img).convert("RGB").resize(img_size)
        photo_path = f"./raw_dataset/photos/photo_{i+1}.jpg"
        photo.save(photo_path)
        painting =
photo.filter(ImageFilter.CONTOUR).filter(ImageFilter.SMOOTH_MORE)
        painting_path = f"./raw_dataset/paintings/painting_{i+1}.jpg"
        painting.save(painting_path)

def preprocess_images(input_folder, domain, resize=img_size):
    images = sorted(Path(input_folder).glob("*.jpg"))
    total = len(images)
    print(f"Processing {total} images from {input_folder}")

    if total == 0:
        print(f"No images found in {input_folder}. Skipping.")
        return

    train_count = int(total * train_split)
    for i, img_path in enumerate(images):
        try:
            with Image.open(img_path) as img:
```

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                img = img.convert("RGB").resize(resize)
                img_name = f"{i:04d}.jpg"
                split = "train" if i < train_count else "test"
                out_folder = os.path.join(output_dir, f"{split}
{domain}")
                os.makedirs(out_folder, exist_ok=True)
                img.save(os.path.join(out_folder, img_name))
        except Exception as e:
            print(f"Error processing {img_path}: {e}")

def analyze_dataset():
    print("\n--- Dataset Analysis ---")
    for folder in ["trainA", "trainB", "testA", "testB"]:
        path = Path(output_dir) / folder
        images = list(path.glob("*.jpg"))
        print(f"{folder}: {len(images)} images")
        if images:
            img = Image.open(images[0])
            print(f"Sample image size in {folder}: {img.size}")

def visualize_samples():
    print("\n--- Sample Images ---")
    folders = ["trainA", "trainB"]
    fig, axs = plt.subplots(2, 5, figsize=(15, 6))
    for i, folder in enumerate(folders):
        path = Path(output_dir) / folder
        images = list(path.glob("*.jpg"))
        for j in range(5):
            axs[i][j].axis('off')
            if j < len(images):
                img = Image.open(images[j])
                axs[i][j].imshow(np.array(img))
                axs[i][j].set_title(f"{folder}/{images[j].name}")
            else:
                axs[i][j].set_title(f"{folder}: n/a")
    plt.tight_layout()
    plt.show()

def run_pipeline():
    create_dir_structure()
    save_sample_images()
    preprocess_images("./raw_dataset/photos", domain="A")
    preprocess_images("./raw_dataset/paintings", domain="B")
    analyze_dataset()
    visualize_samples()

if __name__ == "__main__":
    run_pipeline()
```

```
Saving sample images...
Processing 5 images from ./raw_dataset/photos
Processing 5 images from ./raw_dataset/paintings

--- Dataset Analysis ---
trainA: 4 images
Sample image size in trainA: (256, 256)
trainB: 4 images
Sample image size in trainB: (256, 256)
testA: 1 images
Sample image size in testA: (256, 256)
testB: 1 images
Sample image size in testB: (256, 256)

--- Sample Images ---
```